Geocomputation

Raster Data Analysis



UCL

Module outline

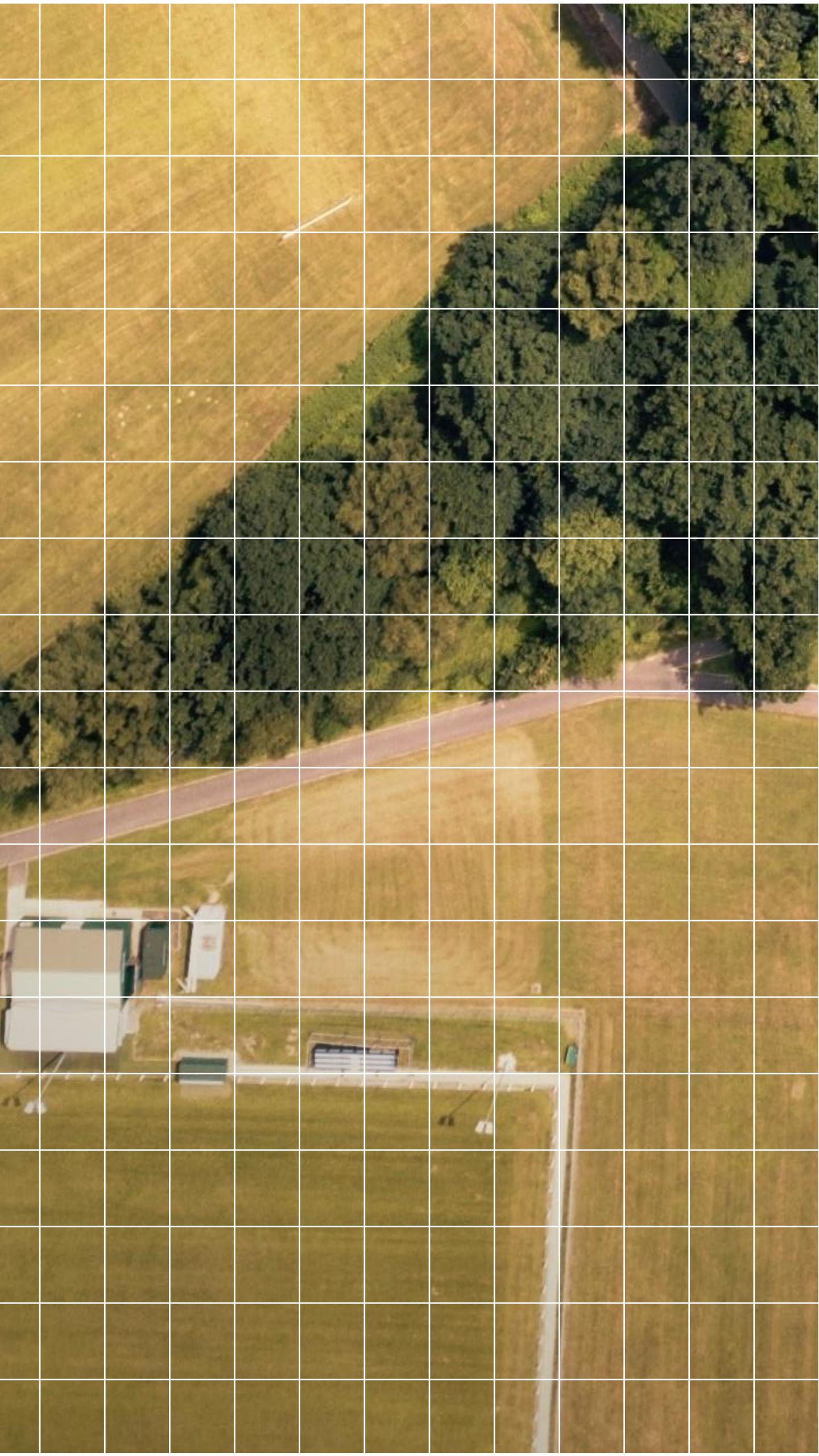
W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

This week

- Raster GIS data model.
- Raster-specific functions and operations.
- Spatial data interpolation.

Before we start – CMD

- Go to www.menti.com
- Use code: XXXX XXXX



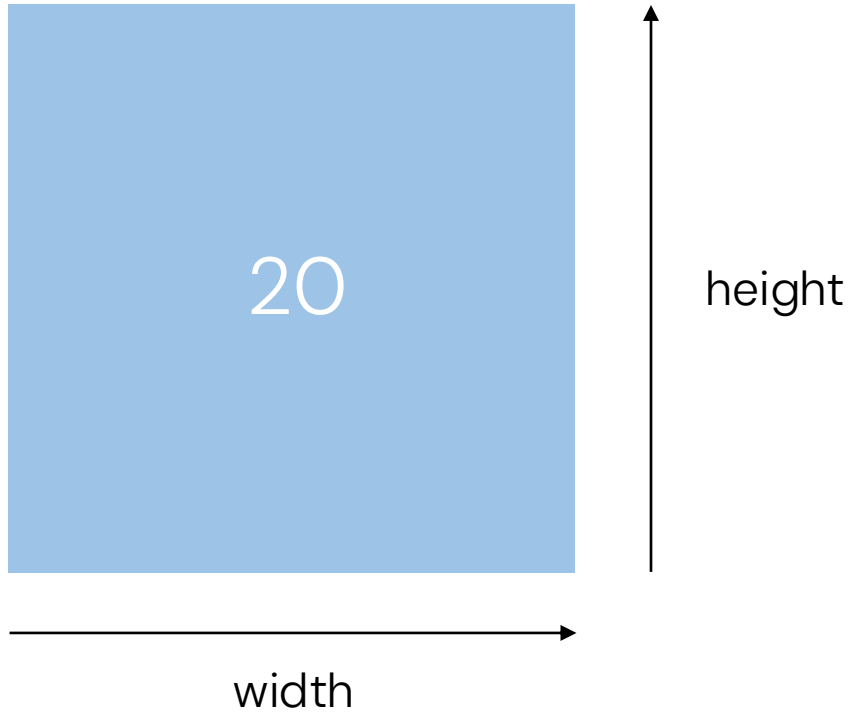


8	9	9	10	0	10	10	10	0	0	0	7	5	3	0	0	0	0	1
8	9	9	10	10	0	10	9	9	0	0	5	3	0	0	0	0	0	0
8	8	9	9	10	0	0	9	8	7	5	0	0	0	1	0	0	0	0
5	8	8	9	10	10	0	9	7	5	0	0	5	5	5	0	0	0	1
3	5	8	9	9	10	0	0	3	0	0	0	5	0	0	1	0	0	2
2	5	8	8	9	9	10	0	0	0	1	5	0	0	0	0	0	0	1
2	4	6	8	8	9	0	0	0	1	5	0	0	5	5	5	0	0	1
0	3	6	8	8	0	0	0	0	5	0	5	5	5	5	5	0	0	0
2	2	5	8	0	0	0	0	0	0	5	5	5	5	5	5	3	0	0
0	2	5	0	0	1	2	3	4	4	4	4	4	4	4	5	0	0	0
0	0	0	0	1	1	1	1	4	4	4	4	4	4	4	5	0	0	0
0	0	1	1	2	2	2	2	3	3	3	3	3	3	3	4	0	3	0
1	1	1	1	2	2	3	3	3	3	1	1	1	1	1	2	3	4	3

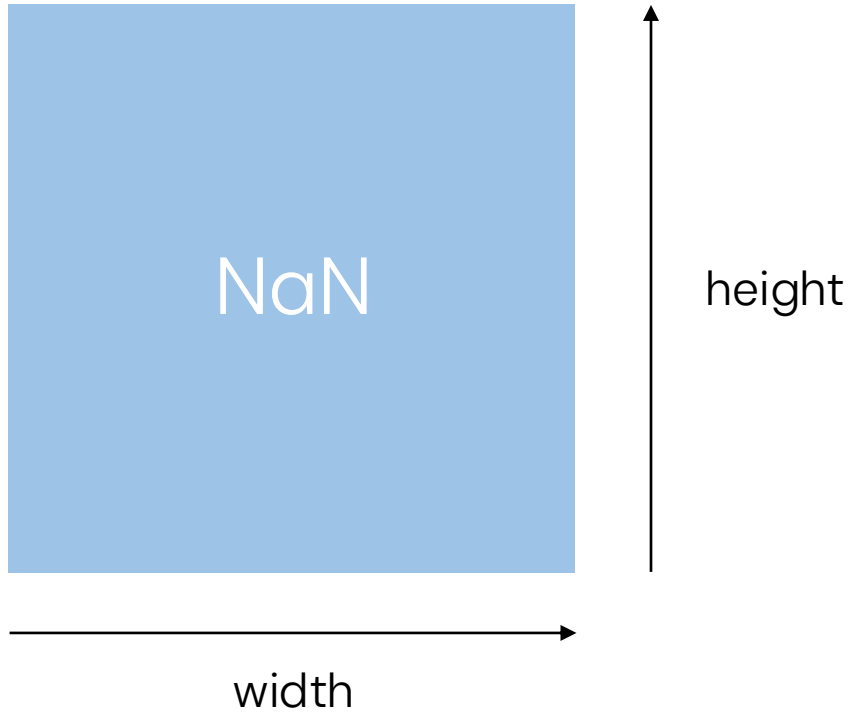
Raster data: a grid of values

- Unlike vector data, raster data can represent features as continuous surfaces, showing gradients of change rather than discrete measurements.
- Raster data is organised as a matrix of pixels or grid cells, each containing a numeric or textual value representing a feature.
- It consists of rows and columns, with each pixel or grid cell defined by its resolution (height and width).

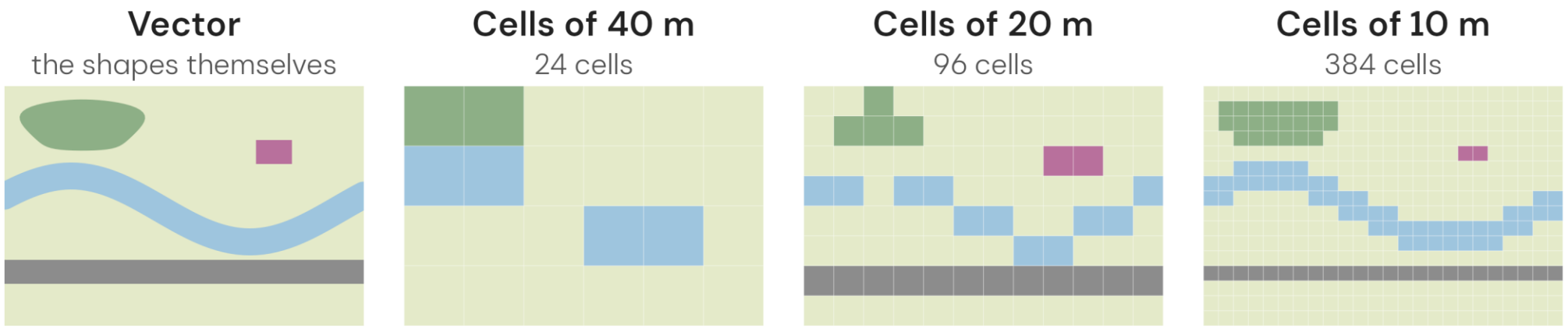
Each cell has a size and a value



Cells without data are NaN



Smaller cells give more detail



Smaller cells give a finer resolution and more detail: here the road and the building appear only once the cells are small enough. Halving the cell size quadruples the cells.

Vector versus raster

- Why not vectorise everything? After all, we can add as many attributes as needed?
- “Raster is faster, but vector is corrector" highlights the trade-offs between these data models.
- The choice between raster and vector data depends on your specific application and the type of analysis you intend to perform.

Where raster data are used

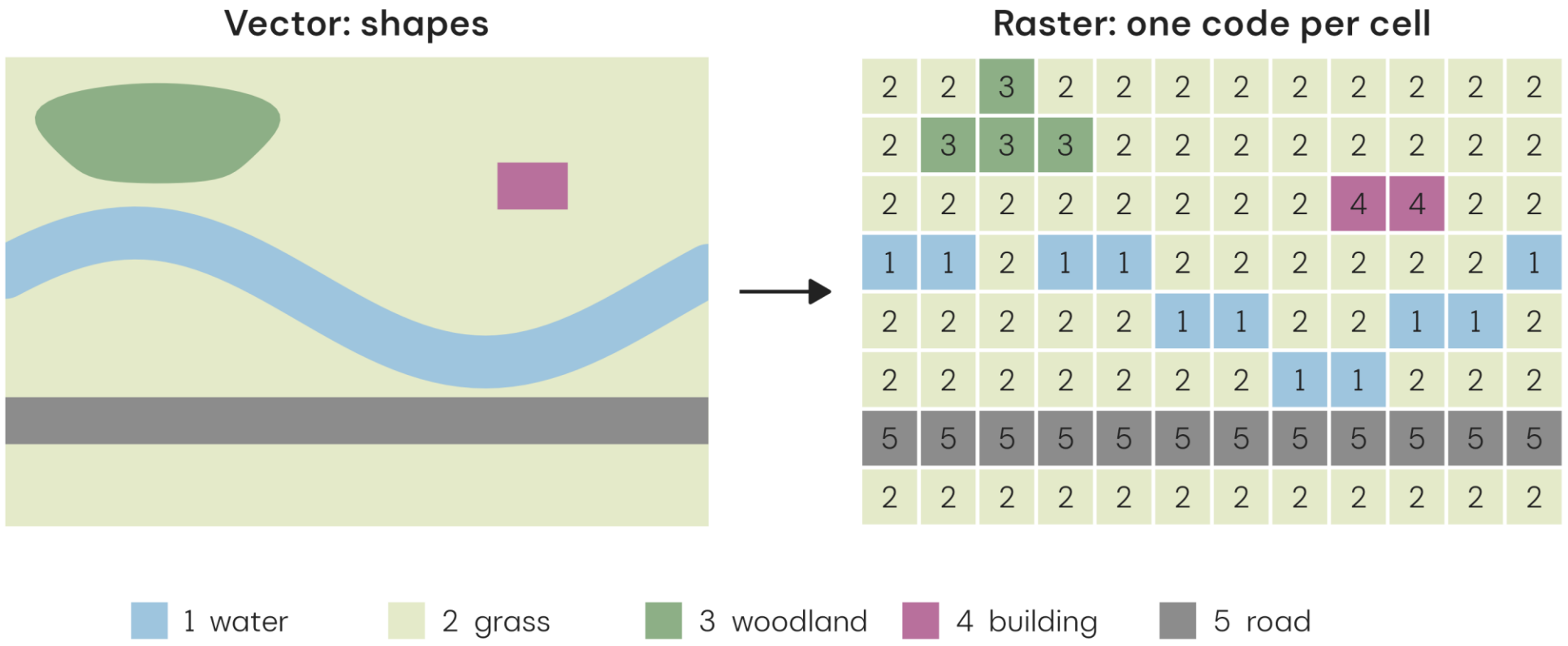
- Raster data are commonly found in applications where every location has one value.
- Variable can be continuous (temperature, elevation, air pollution), categorical (land use), or imagery (reflectance values).
- Cell size is constant across the raster surface.
- When do we typically encounter raster data:
  - Remote sensing
  - Outcome of spatial analysis

Raster versus vector: trade-offs

- Raster
  - Quick and easy map algebra, and the only option for some analyses, such as water flow over a land surface.
  - Pixelated look, linear features are hard to show, and files get large because every cell holds a value.
- Vector
  - Precise shapes and attractive maps; accuracy does not depend on a grid size.
  - Continuous data are poorly stored and displayed, and the data need a lot of maintenance to stay accurate.

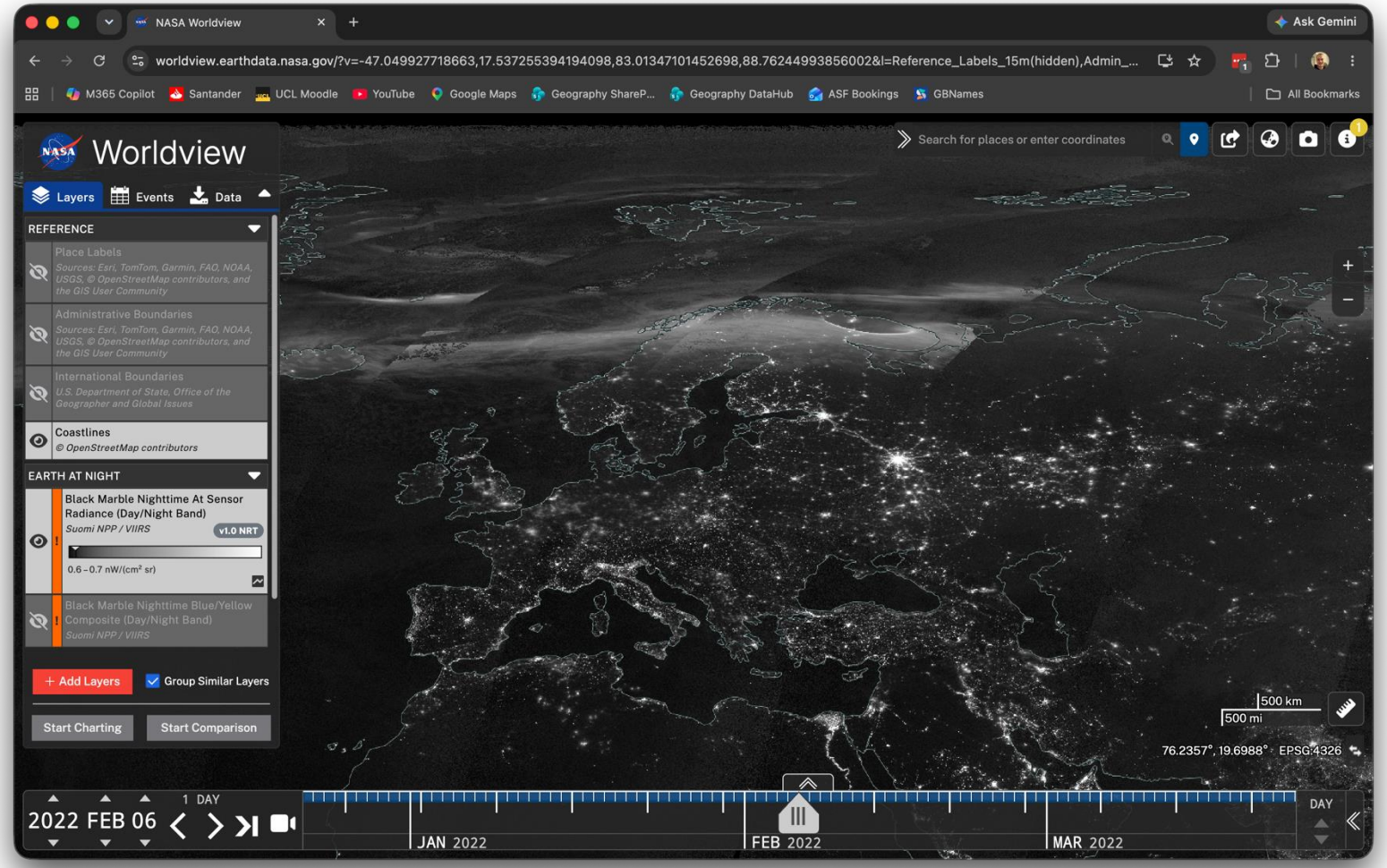
Adapted from a comparison by CARTO.

Categorical rasters: values are codes

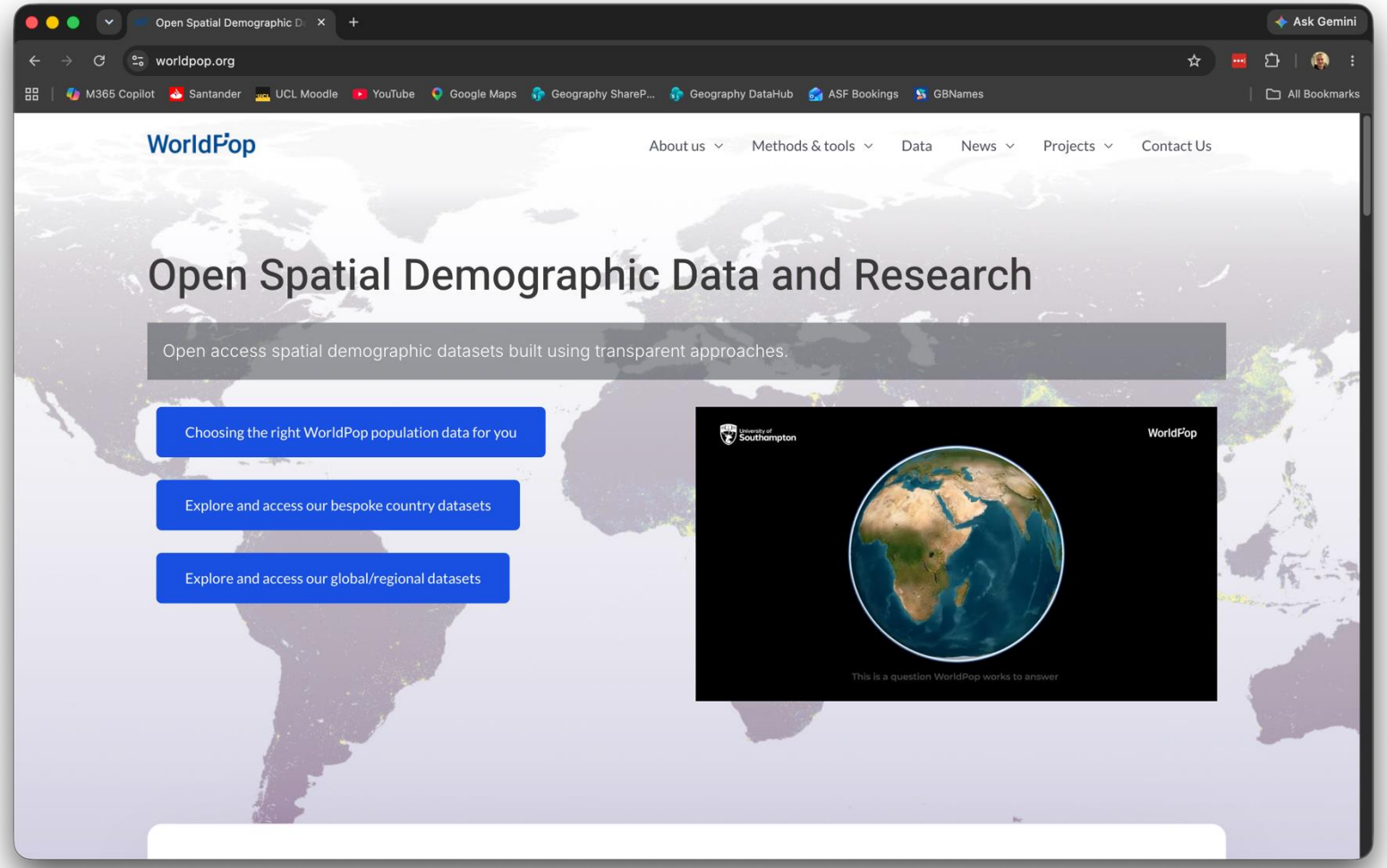


The codes are categories, not amounts.

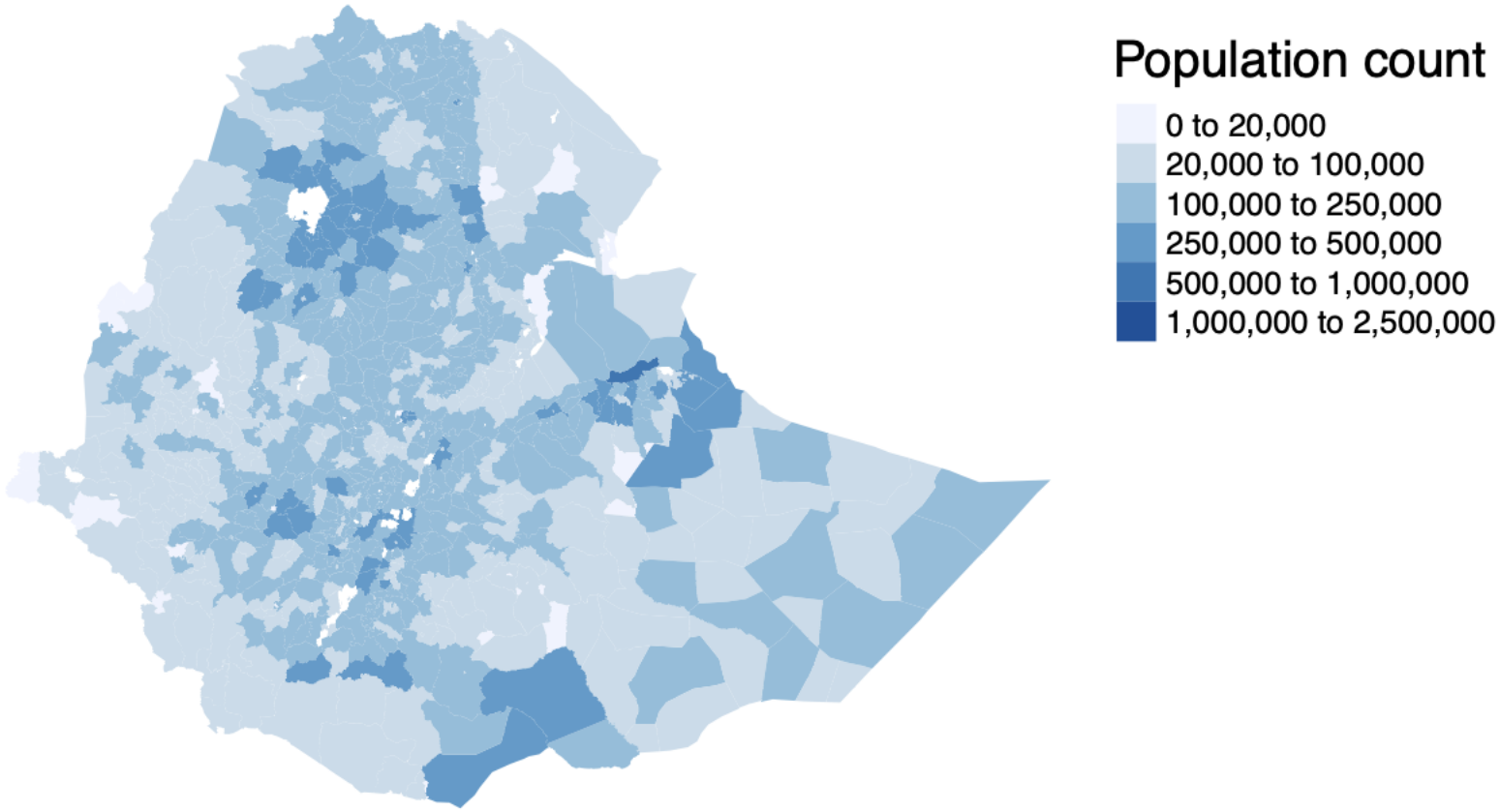
Continuous rasters: values on a scale



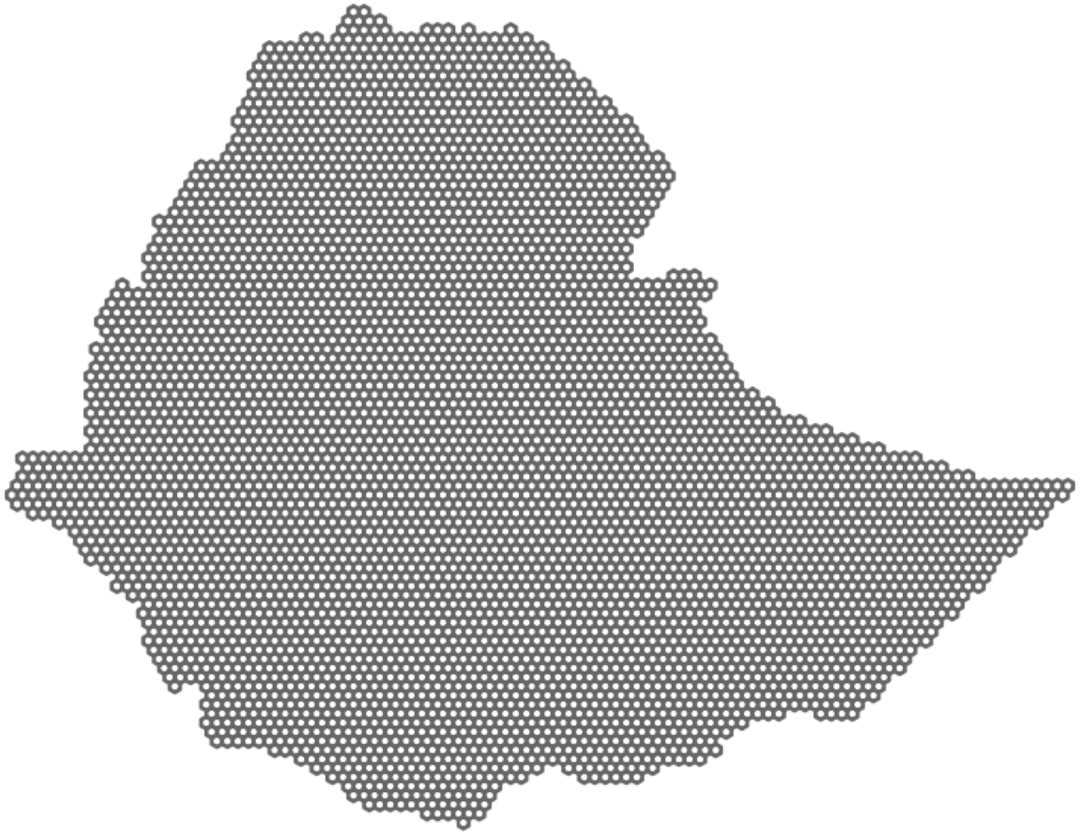
WorldPop: population on a grid



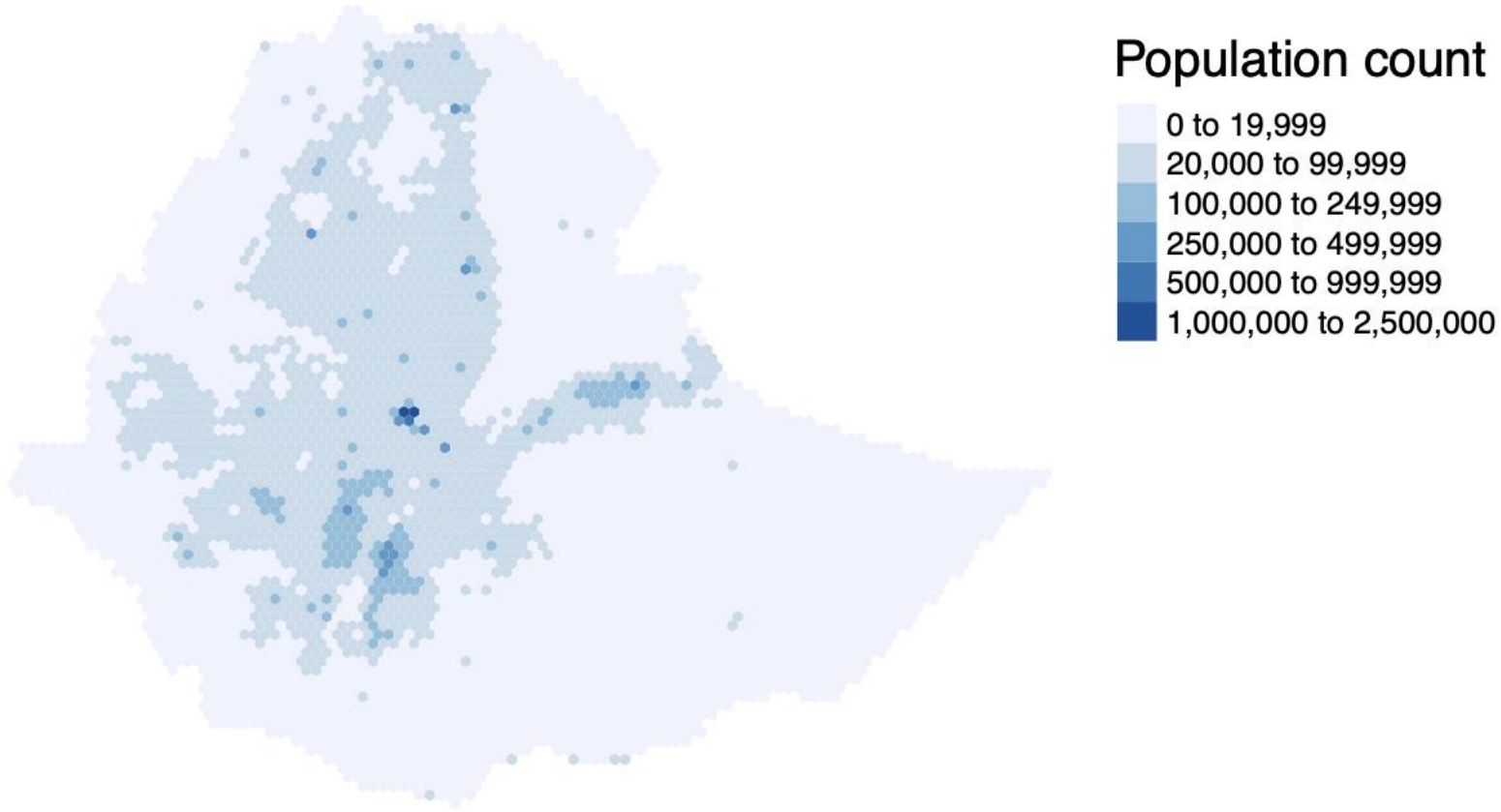
Population counts by administrative area



The same area as a grid



Population counts per grid cell



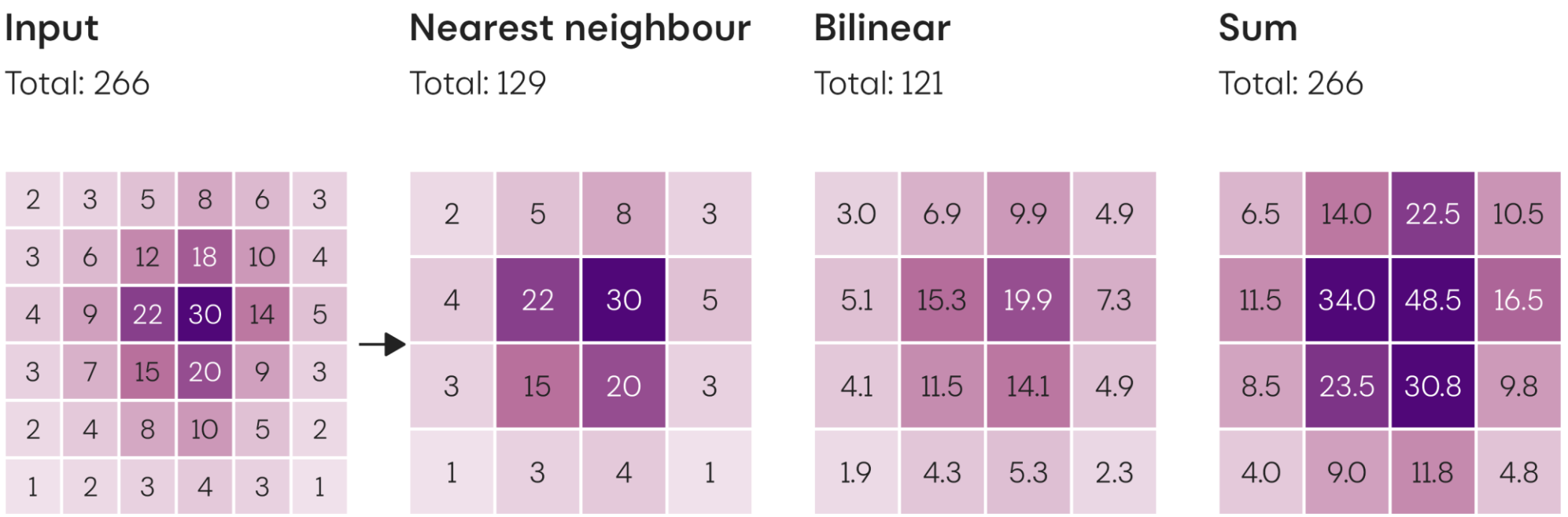
Raster files

- Multiple data formats, including GeoTIFF, ASCII Grid, Esri Grid and GIF.
- Must be georeferenced (i.e. have a CRS) to plot properly.
- Using raster data also means we should use raster-specific functions and operations.

Combining rasters: same grid first

- To combine rasters cell by cell, they need the same coordinate reference system, cell size and extent.
- If they differ, reproject or resample one raster to match the other.
- The resampling method depends on the data: nearest neighbour for categories, bilinear for continuous values, sum for counts.

Resampling: match the method to the data

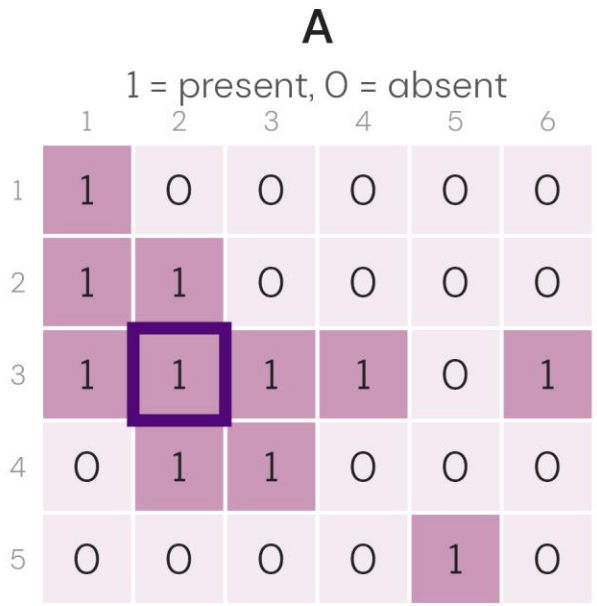


Only the sum keeps the total number of people the same, so use it for a raster of counts.

Four types of operation

- Local: each cell on its own, or the same cell in several rasters.
- Focal: a cell and its neighbours.
- Zonal: all cells in the same zone.
- Global: all cells in the raster.

A raster is a matrix of values



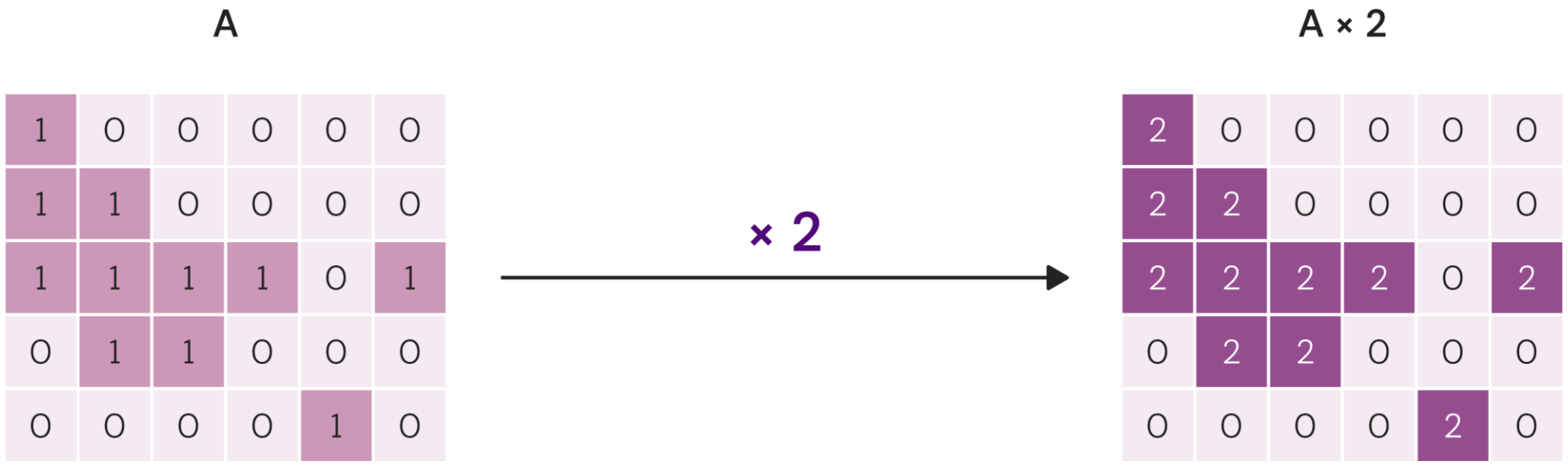
A raster is a grid of rows and columns.

Every cell holds one value.

The outlined cell, row 3 and column 2, holds a 1.

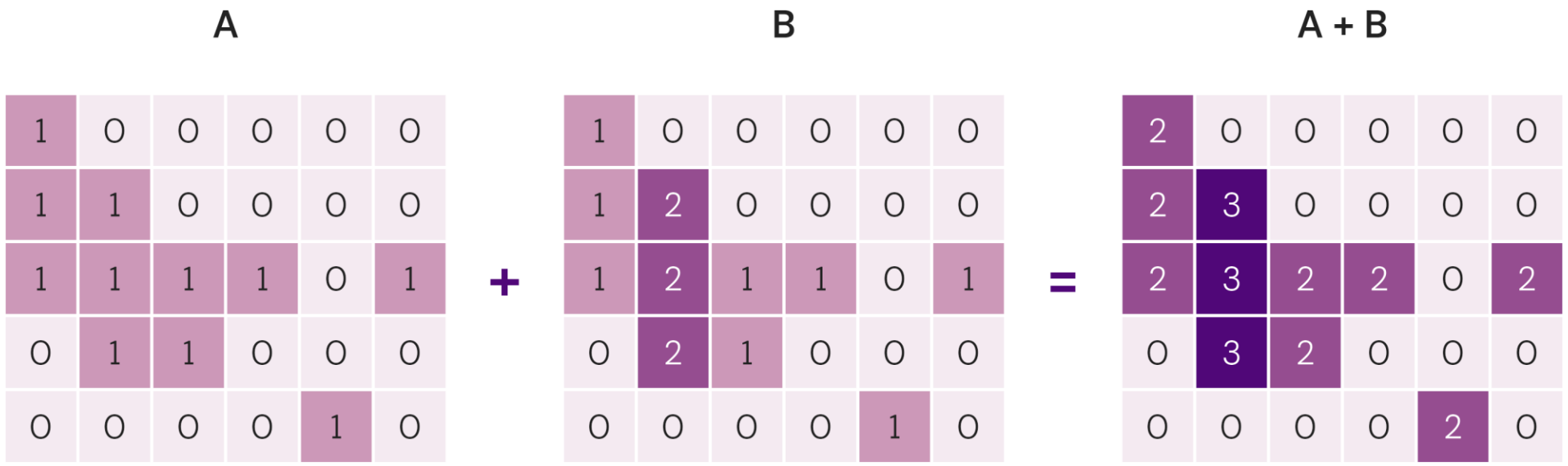
Local operations and functions

Local: arithmetic on one raster



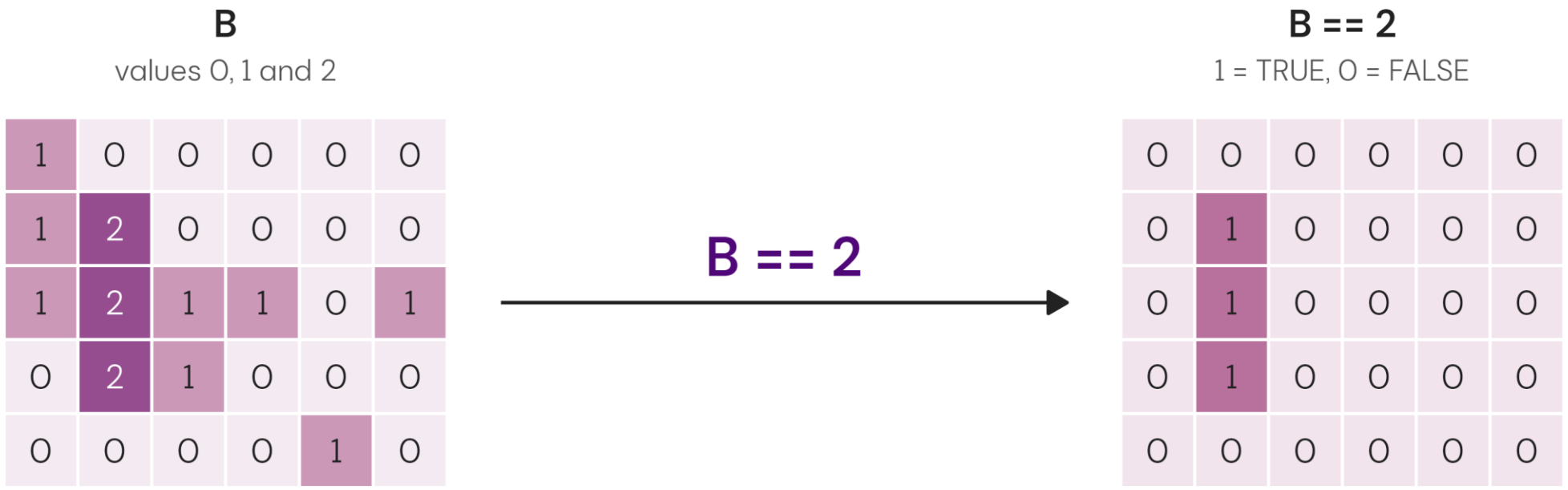
Every cell is multiplied by 2 on its own: no other cell is involved.

Local: arithmetic on two rasters



Cells in the same position are added. The rasters need the same grid.

Local: comparison gives TRUE or FALSE



A comparison asks a question of every cell and returns TRUE or FALSE.

Boolean AND: both must be true



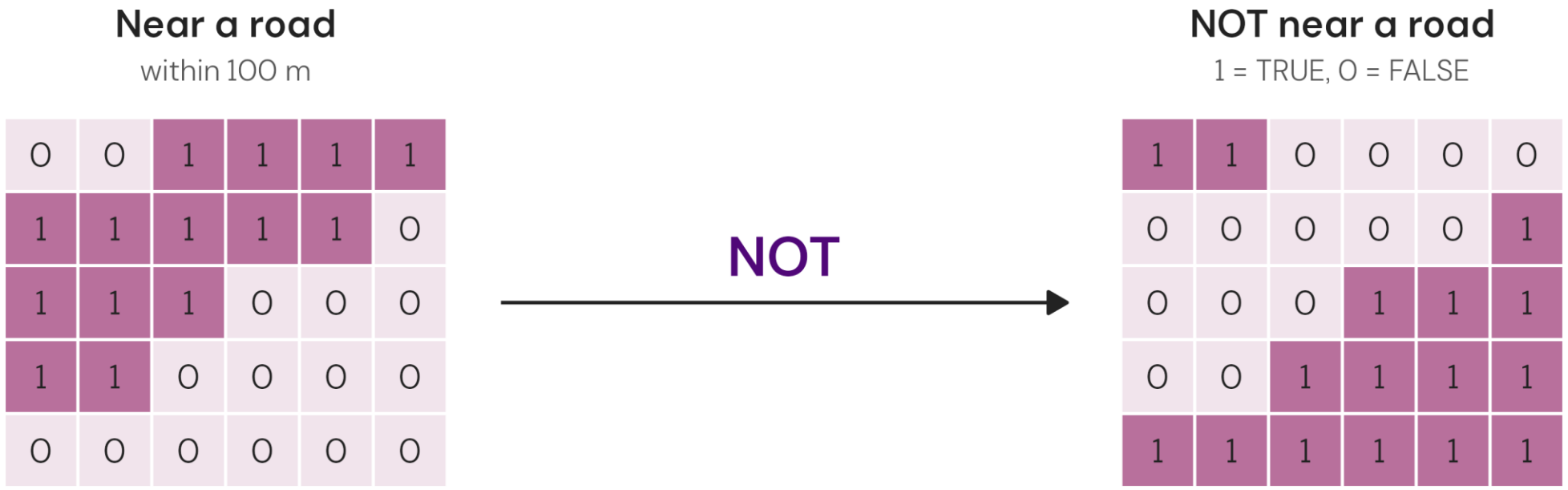
AND: a cell is TRUE only if both layers are TRUE.

Boolean OR: at least one is true



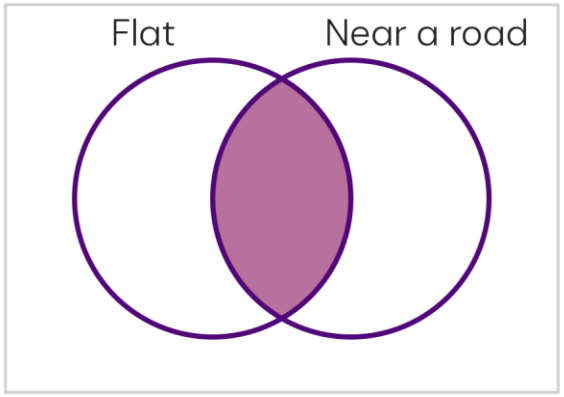
OR: a cell is TRUE if at least one of the layers is TRUE.

Boolean NOT: reverse a layer



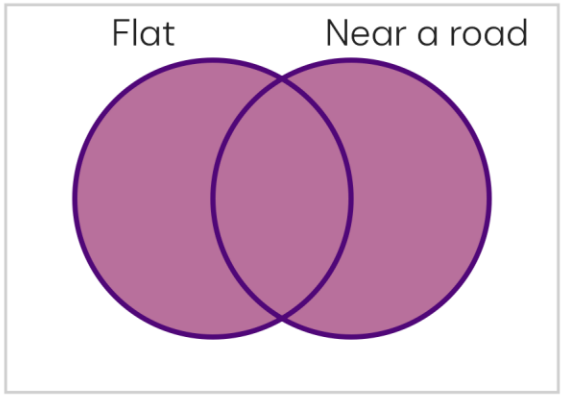
NOT reverses a layer: TRUE becomes FALSE and FALSE becomes TRUE.

AND, OR and NOT at a glance



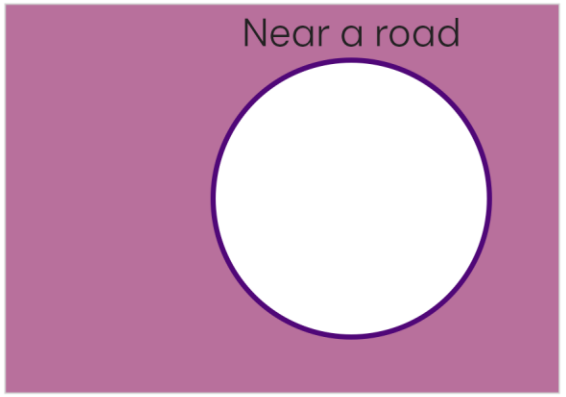
AND

both layers are TRUE



OR

at least one is TRUE



NOT

reverses a layer

The shaded area is TRUE. The box is every cell of the raster.

Focal operations and functions

Focal: sum a window around a cell



The window holds the cell and its neighbours. In the corner it has 4 cells: 1 + 0 + 1 + 2 = 4.

The window moves to the next cell



The window moves one cell down. At the edge it has 6 cells: 1 + 0 + 1 + 2 + 1 + 2 = 7.

In the middle the window is full



In the middle the window has 9 cells: 2 + 0 + 0 + 2 + 1 + 1 + 2 + 1 + 0 = 9.

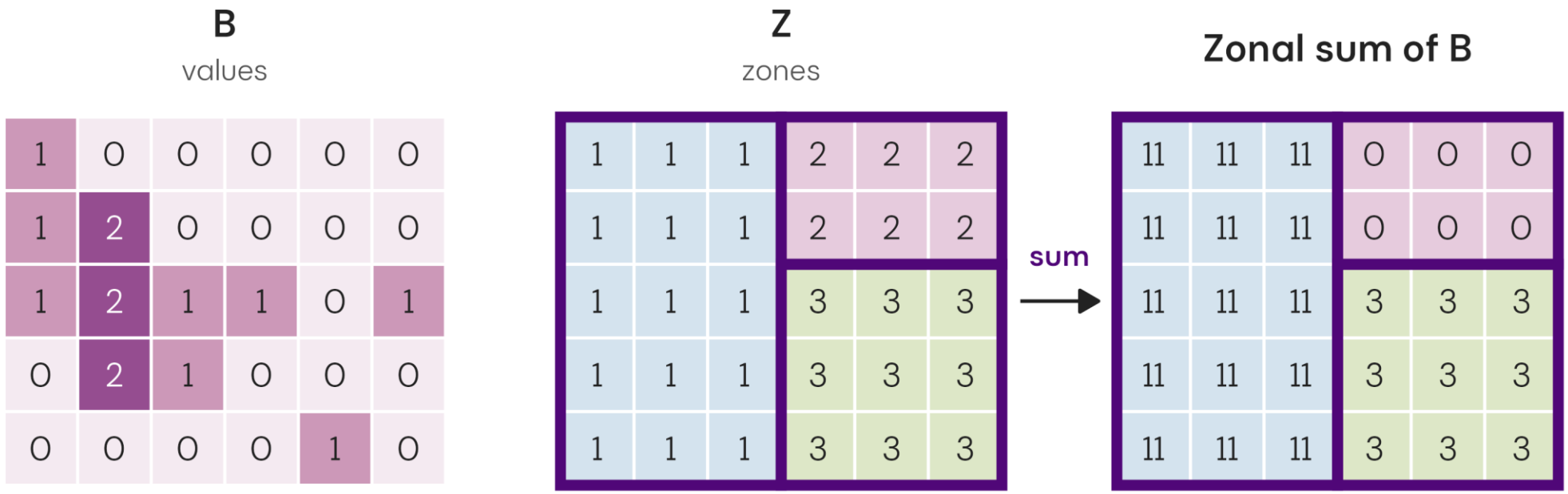
The result: a summary for every cell



The window visits every cell: each cell of the result summarises its neighbourhood.

Zonal operations and functions

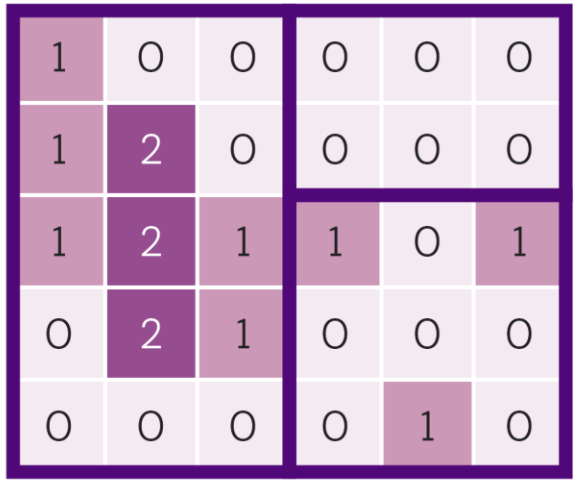
Zonal: summarise the cells in a zone



Every cell takes the sum of B over its zone: zone 1 is 11, zone 2 is 0 and zone 3 is 3.

Zonal statistics with polygons

B with zones



Zonal statistics

Zone	Sum
1	11
2	0
3	3

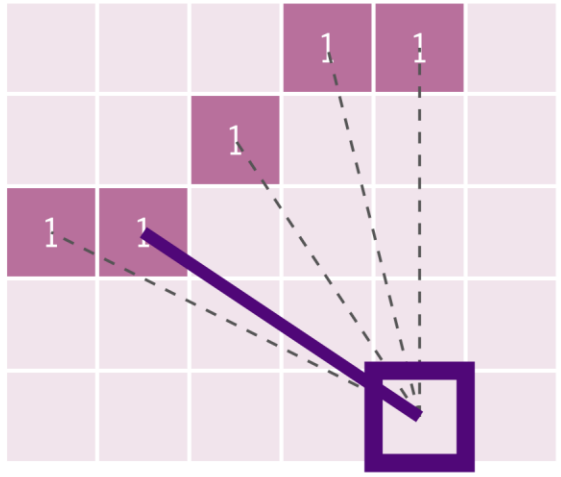
With polygons as zones, the result is a table with one row per zone.

Global operations and functions

Global: distance to the nearest road

Roads

1 = road cell



distance to the nearest road

Distance (m)

cells of 100 m

200	141	100	0	0	100
100	100	0	100	100	141
0	0	100	141	200	224
100	100	141	224	283	316
200	200	224	283	361	412

To find 361 m, the outlined cell compares its distance to every road cell.
Every output cell uses the whole input raster.

Operators combine into a site search

Slope
degrees

8	7	5	3	2	5
7	6	5	2	1	2
5	5	2	1	3	6
3	2	1	2	5	7
2	3	5	6	8	9

and

Distance to road
metres

200	141	100	0	0	100
100	100	0	100	100	141
0	0	100	141	200	224
100	100	141	224	283	316
200	200	224	283	361	412

=

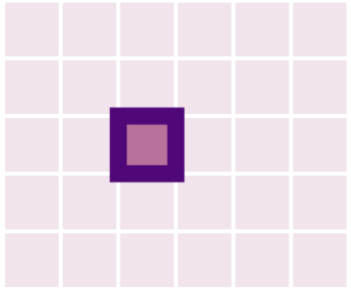
Suitable
slope < 5 AND distance ≤ 100

0	0	0	1	1	0
0	0	0	1	1	0
0	0	1	0	0	0
1	1	0	0	0	0
0	0	0	0	0	0

Operators combine in one expression: 7 cells are flat and near a road.

Four types of operation at a glance

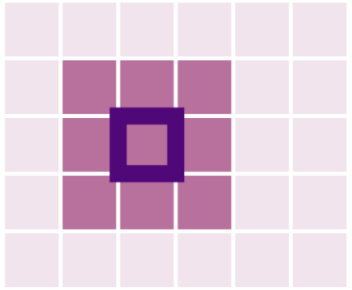
Local



the same cell in each raster

uses 1 cell

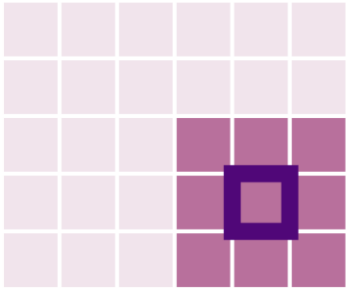
Focal



a cell and its neighbours

uses 9 cells

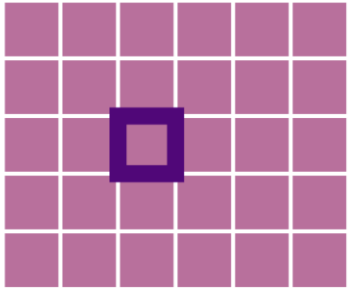
Zonal



all cells in the same zone

uses 9 cells

Global



all cells in the raster

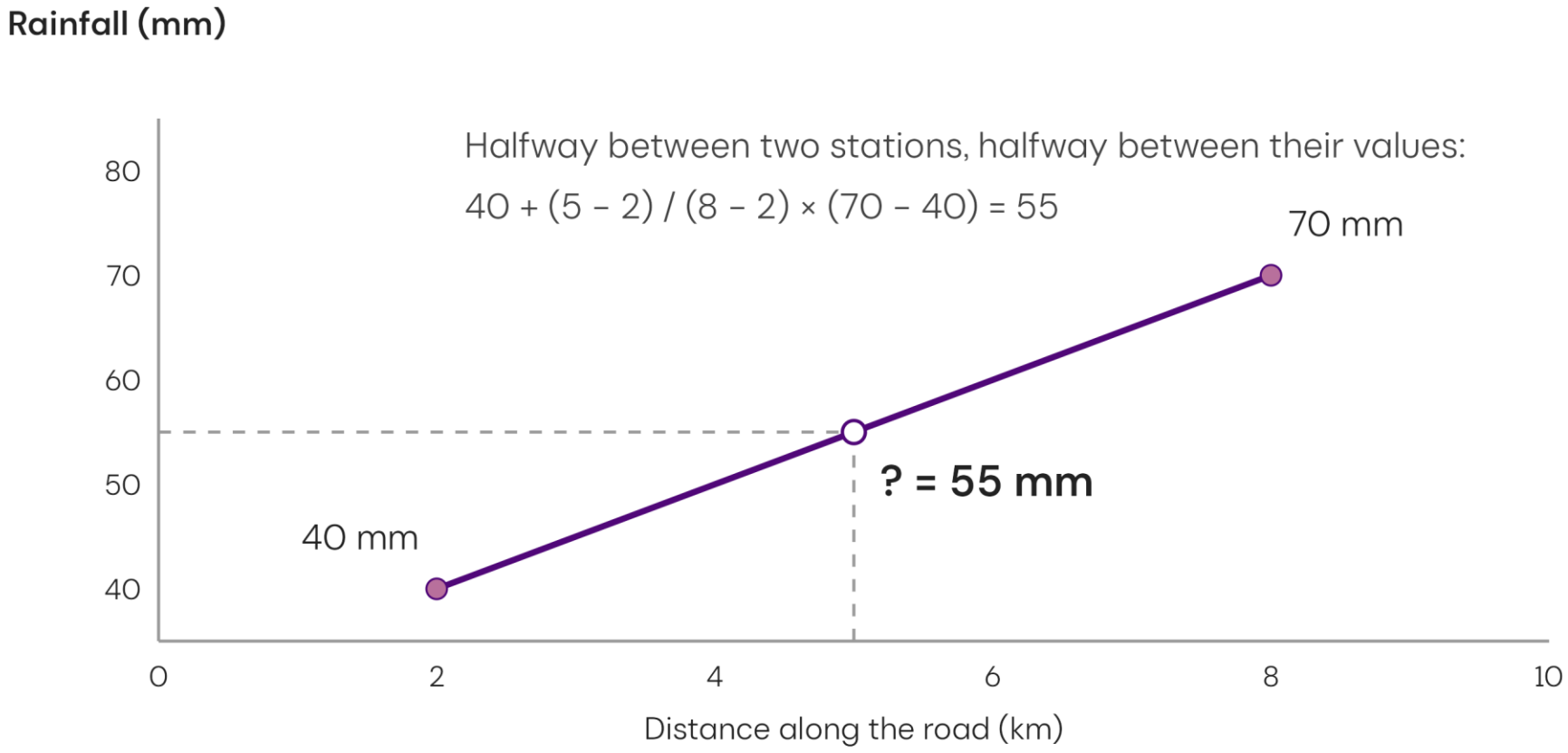
uses 30 cells

cells that feed the result

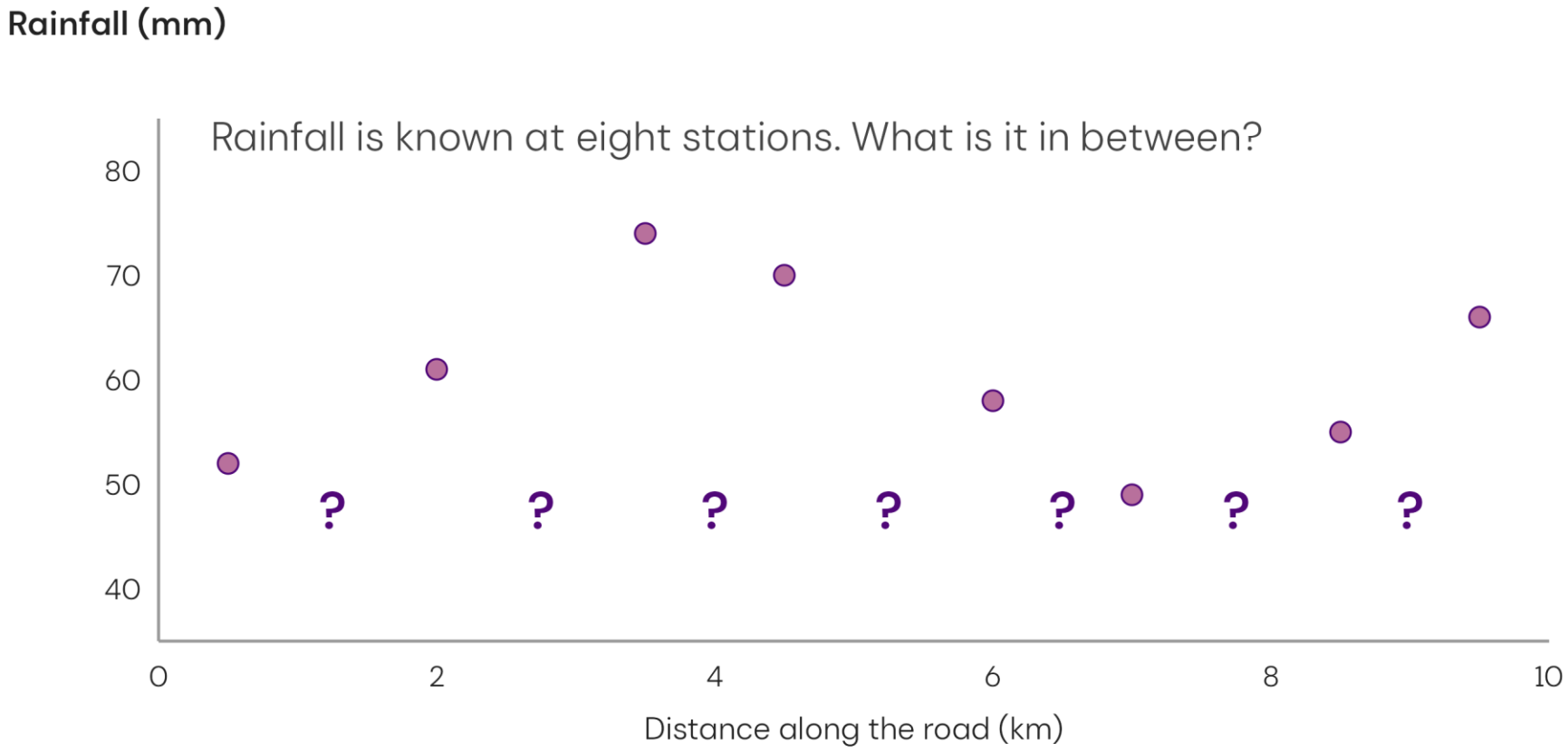
the output cell being calculated

Data interpolation

Interpolation: between known values

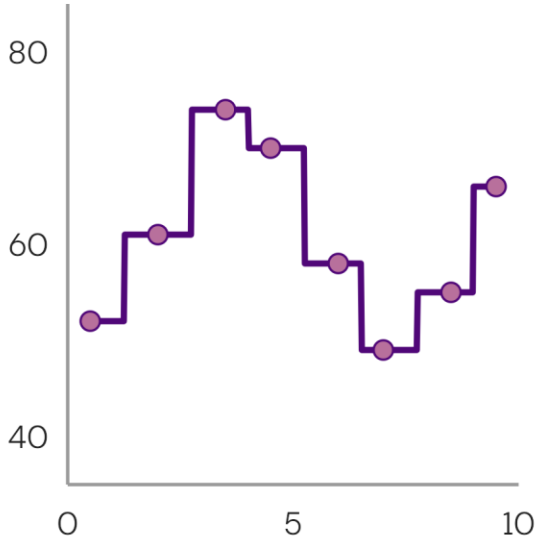


Known values at a few places



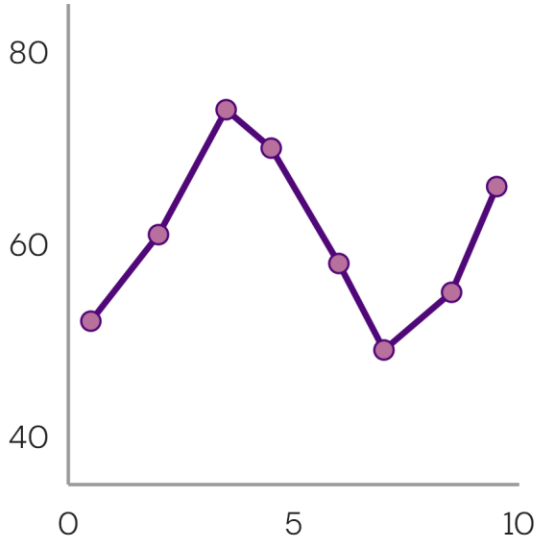
The method decides the estimate

Nearest neighbour



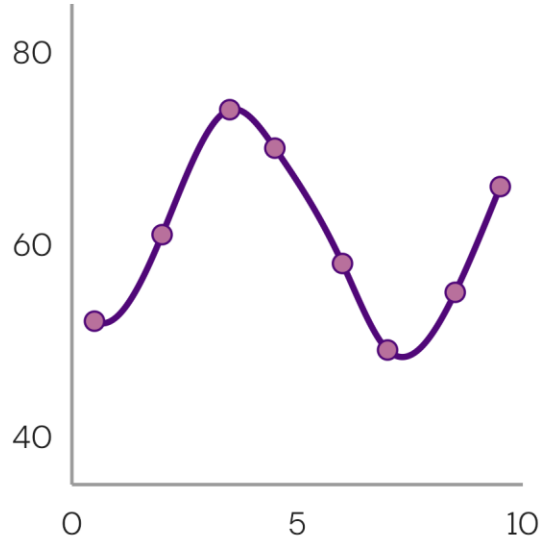
steps: the value of the closest station

Linear



straight lines between the stations

Smooth curve



a smooth curve through the stations

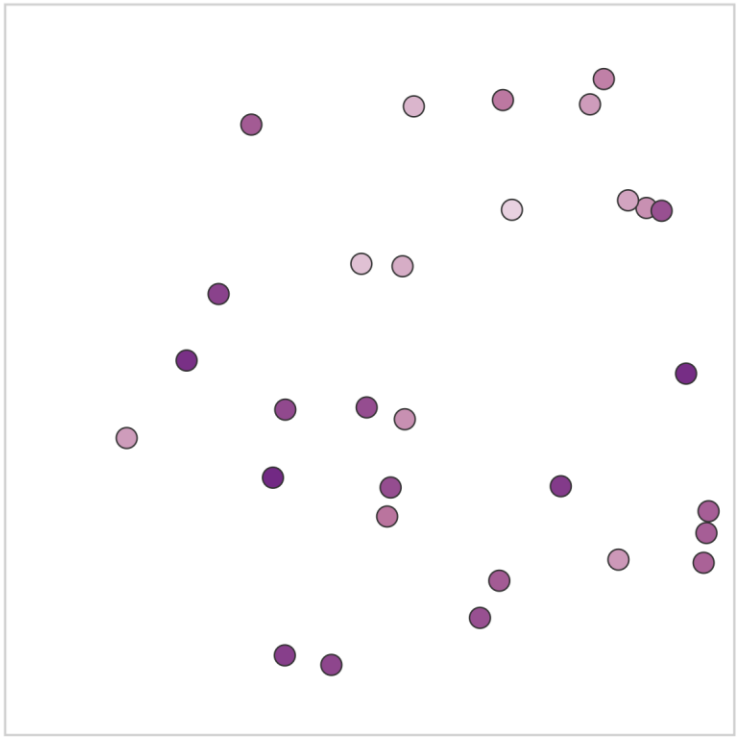
Every method passes through the measured values, but the estimate in between depends on the method.

Spatial interpolation: points to surface

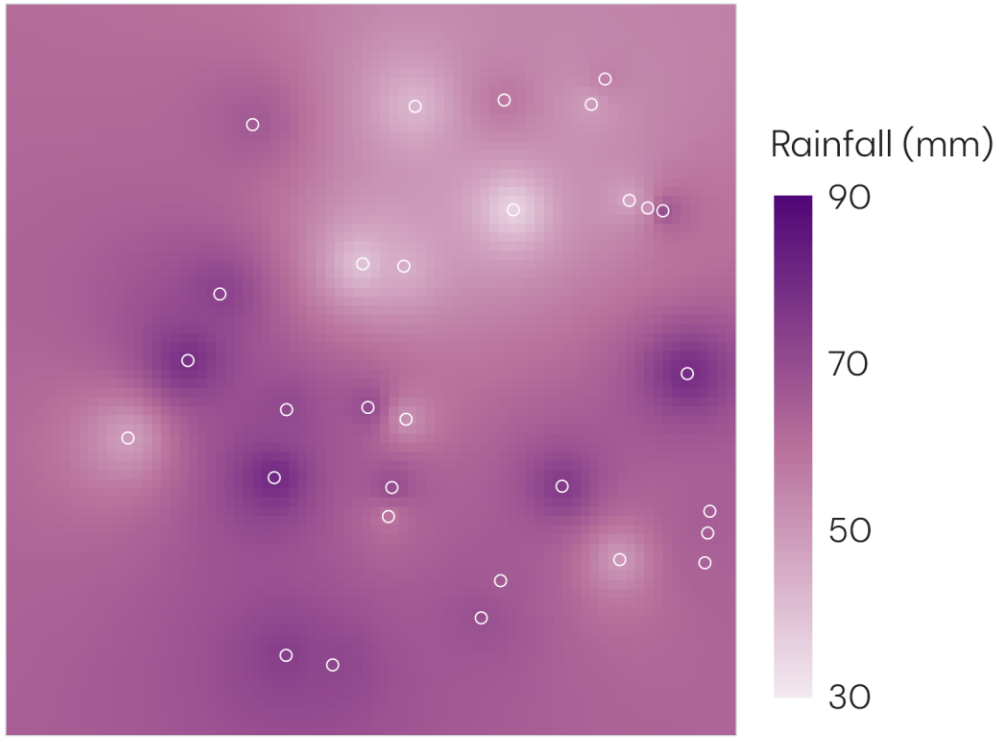
- Similar to 'regular' data interpolation, but in two dimensions, accounting for spatial relationships.
- Commonly used for phenomena that are difficult to measure directly, such as spatial concentrations.
- Uses sampled points with known values to create a continuous (predictive) surface.
- There are various methods available to derive interpolated values, depending on the data and application.

From measured points to a surface

Measured at 30 stations



Interpolated surface



Deterministic and geostatistical methods

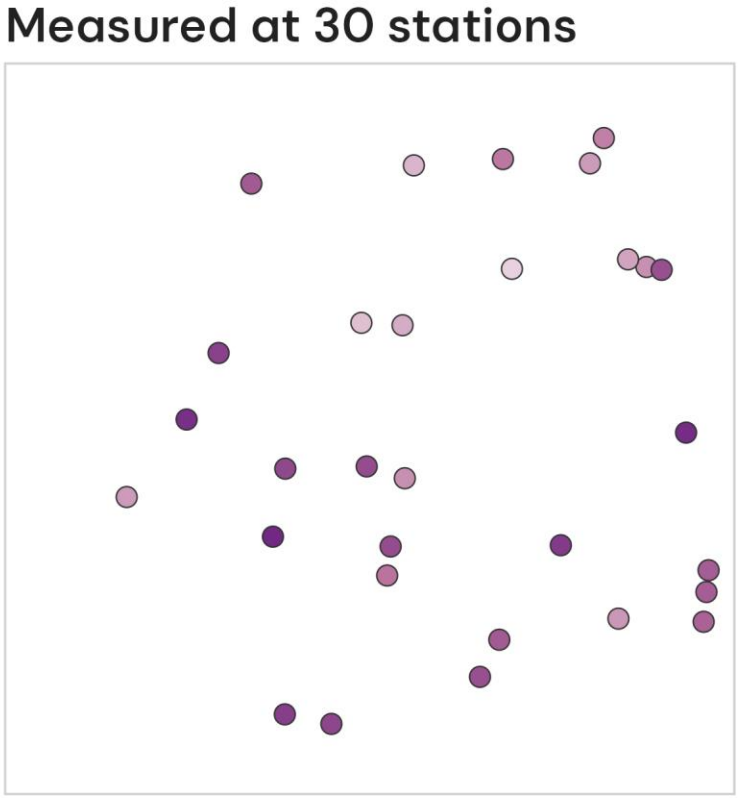
Deterministic methods

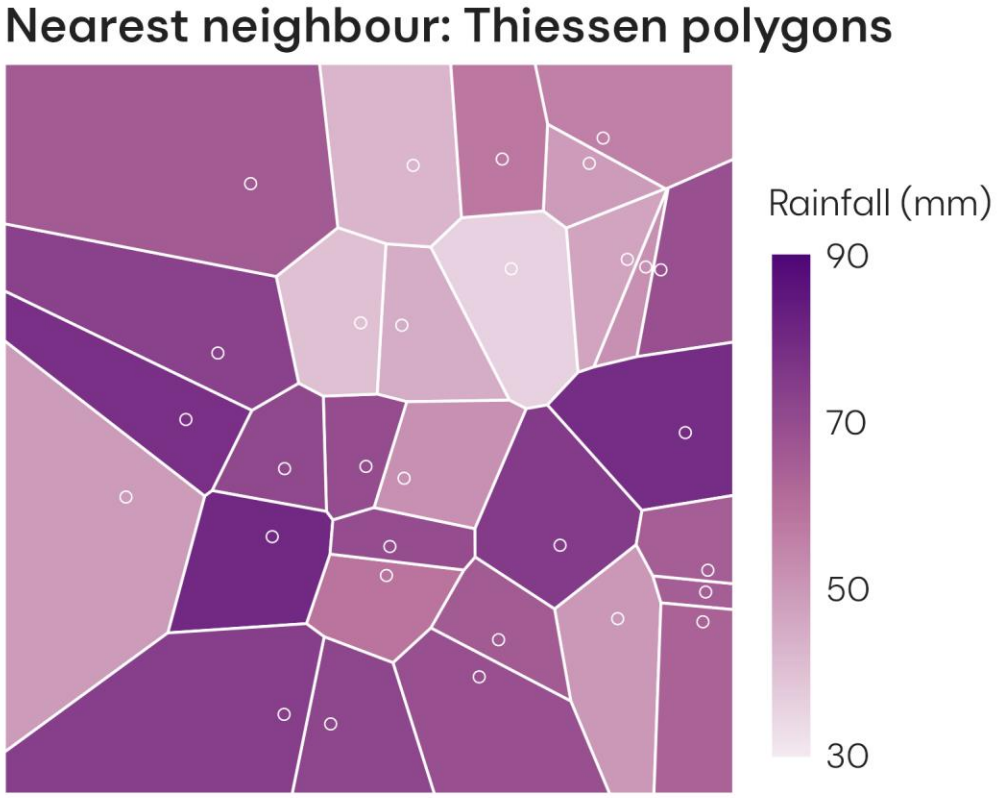
- The user sets the parameters, for example the power in IDW.

Geostatistical methods

- The parameters are estimated from the data, for example in kriging.

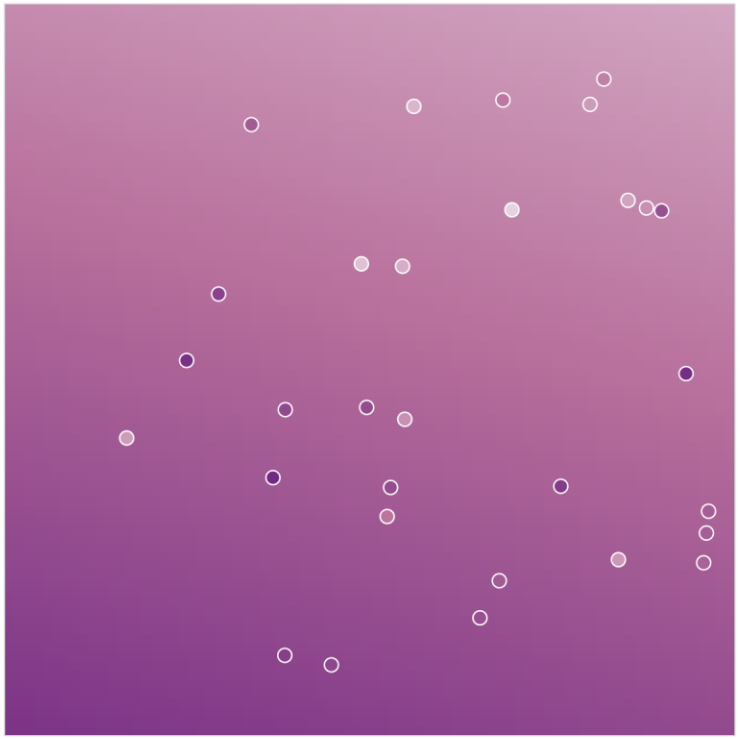
Nearest neighbour: Thiessen polygons





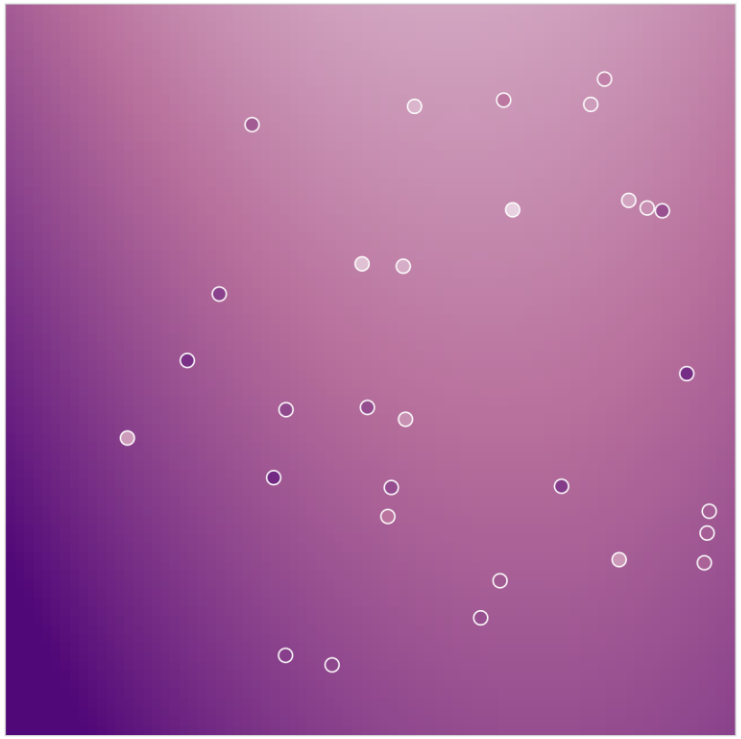
Trend: a surface fitted to all points

First order: a plane

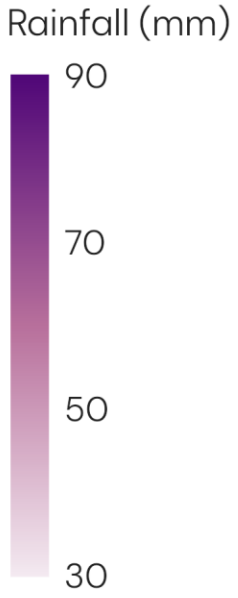


typical gap at the stations: 11 mm

Second order: a curved surface



typical gap at the stations: 10 mm

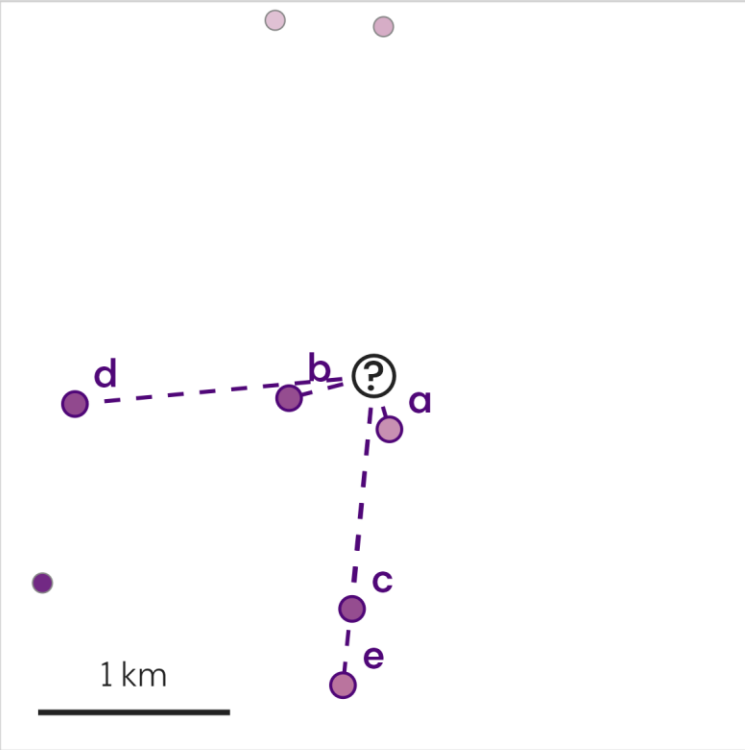


IDW: nearer points get more weight

- IDW: Inverse Distance Weighting gives nearby points more weight.
- Based on Tobler's first law: near points are more alike than far points (distance decay).
- Spatial autocorrelation is the underlying assumption of IDW.
- The power p sets how fast the influence of a point fades with distance.

IDW: an estimate at one location

Zoomed in on the location marked ?



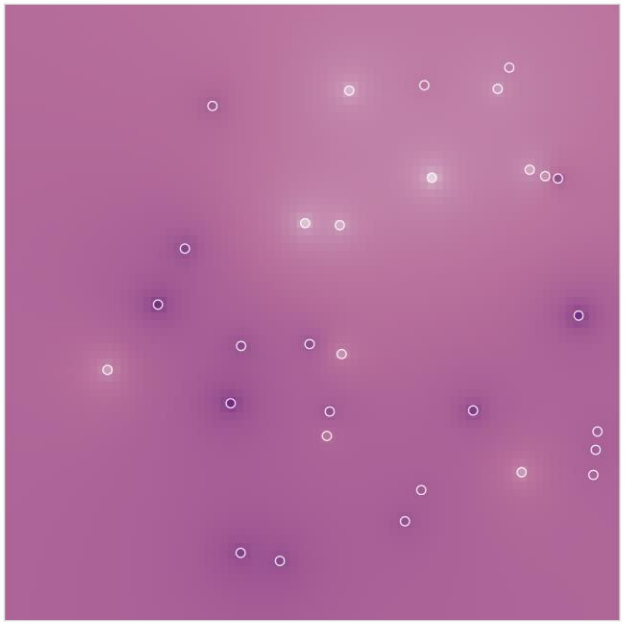
	Rain (mm)	Distance (km)	Weight 1 / d²	Share (%)
a	52	0.29	11.98	66
b	70	0.46	4.79	26
c	70	1.22	0.68	4
d	71	1.56	0.41	2
e	59	1.62	0.38	2

Estimate = sum of (weight × rain) / sum of weights = 58 mm

The plain average of the five stations is 64 mm: the nearest station counts most.

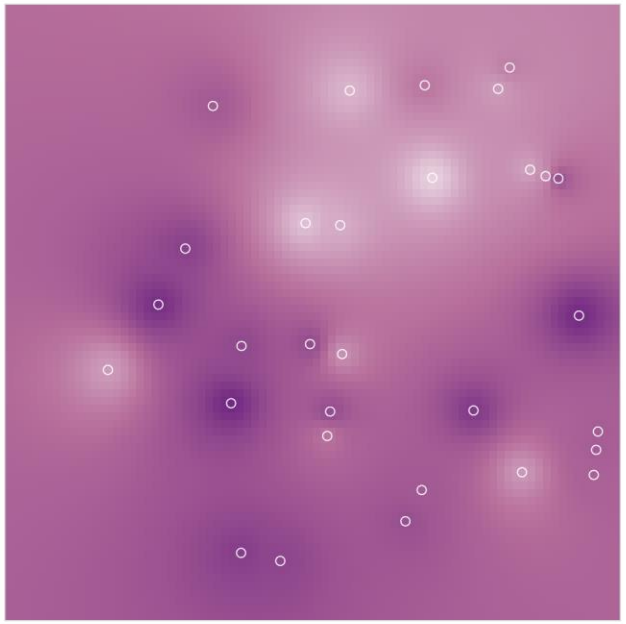
IDW: the power sets the influence

Power 1



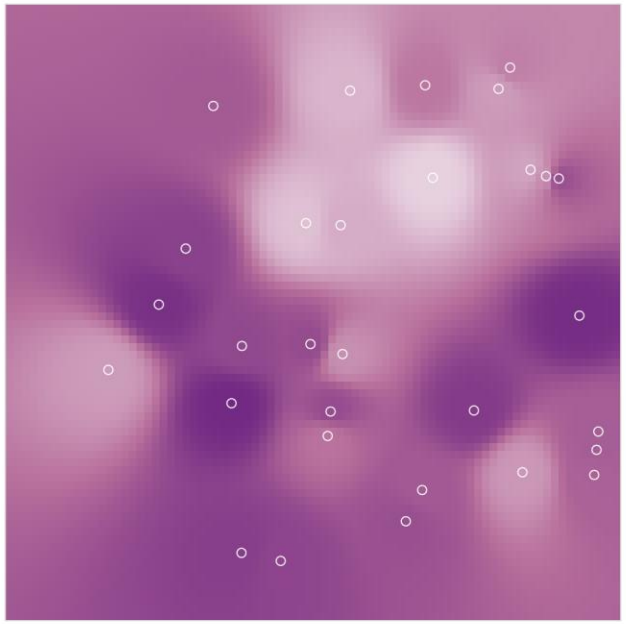
smoother, distant stations count

Power 2



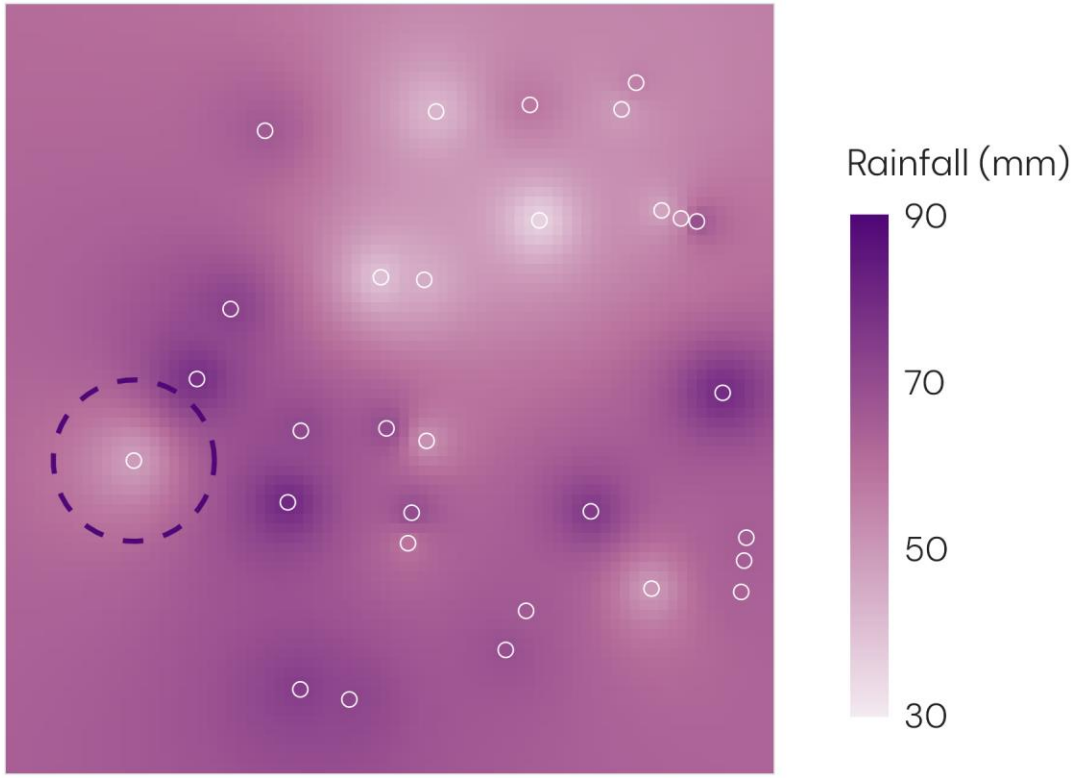
the middle way

Power 4



local, rings around stations

IDW: the result and its limits



Range of the stations

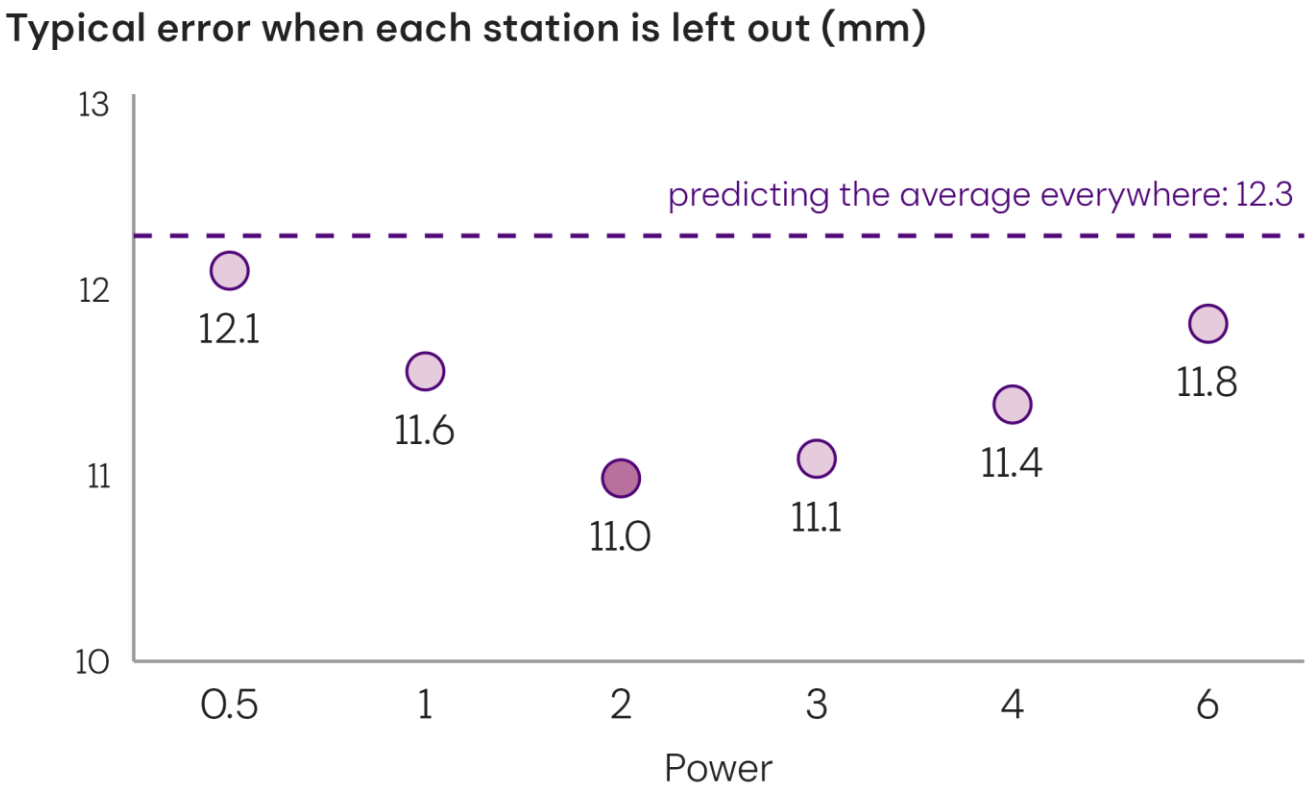
36 to 80 mm

Range of the surface

36 to 80 mm

The surface never goes beyond the measured values and forms rings around stations that differ from their neighbours (dashed ring).

Choosing the power: leave one out



Leave one out:

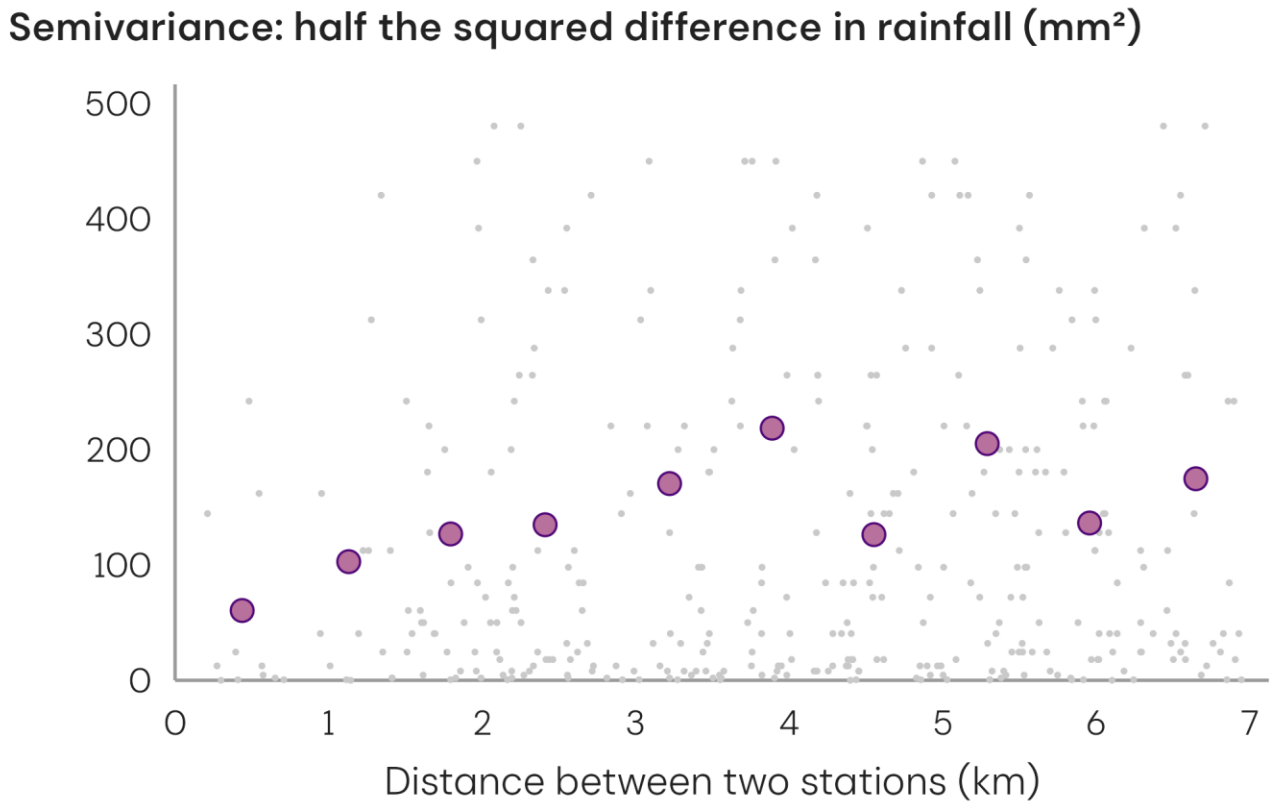
remove a station, predict it from the other 29 and compare with what was measured.

Repeat for every station and for each power. The power with the smallest error is a good choice, not a final answer.

Kriging: estimate how similarity fades

- Considers both distance and degree of variation between known data points when estimating values in unknown areas.
- Kriging is a complex, multistep process starting with a semi-variogram.

Semi-variogram: pairs of stations

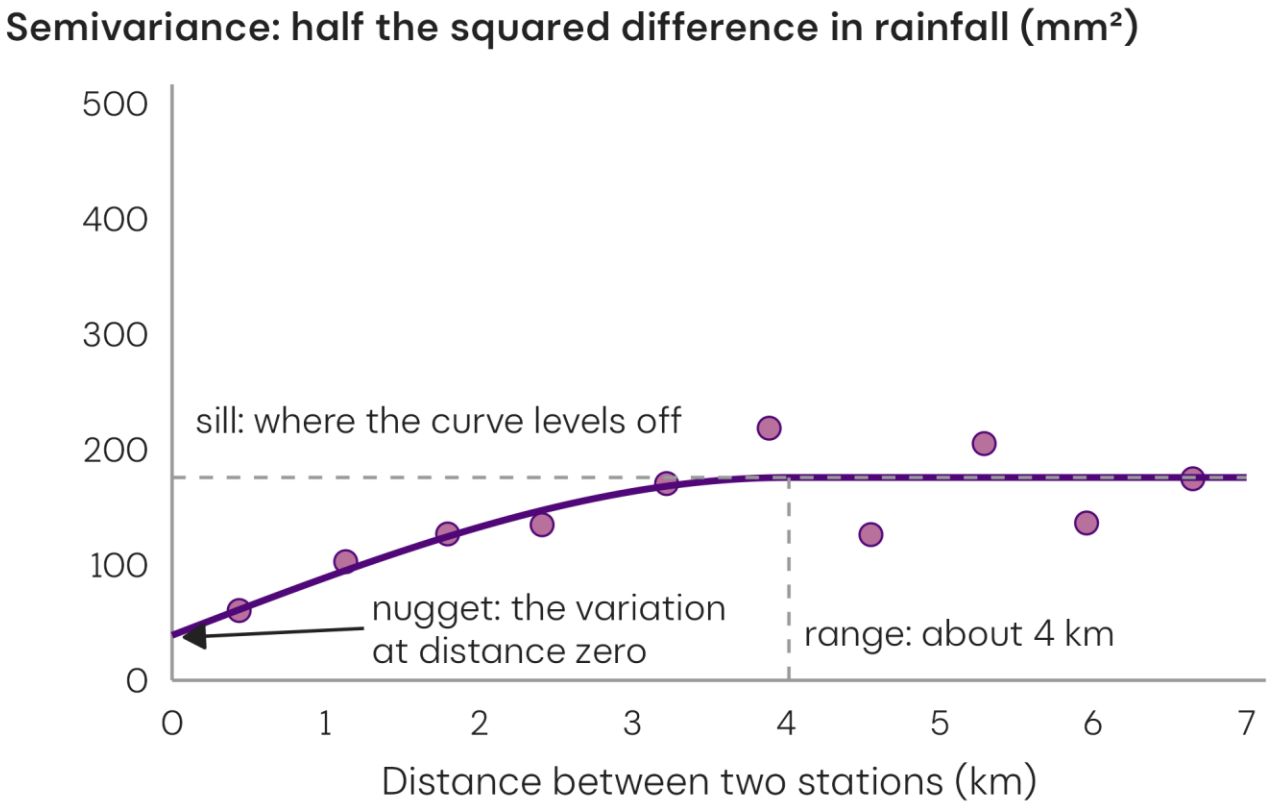


Every pair of stations gives one grey point:

30 stations make
435 pairs.
28 pairs above 500
are off the chart.

average per distance band

Semi-variogram: a fitted curve



A curve is fitted to the averages. Kriging uses it to set the weights.

Stations closer than the range resemble each other; beyond it they do not.

IDW versus kriging: fixed or estimated

- IDW (Inverse Distance Weighting) assumes that spatial autocorrelation between neighbouring points is inversely proportional to the distance between them, and this relationship can be defined by a distance-based function.
- Kriging uses the data to estimate how quickly similarity declines with distance (the semi-variogram) and uses the result to set the weights.

Conclusion

- Raster data model is essential for representing categorical and continuous data.

  Raster data come with their own set of functions and operations.
- Spatial data interpolation is the idea to estimate unknown values of a phenomenon.

  The result is a continuous surface, usually stored as a raster.
- Multiple methods for interpolation (e.g., IDW, Kriging) that vary in assumptions.

Questions

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